

DATA SCHOOL HACKATHON 2026 · STANDARD BANK CORPORATE & INVESTMENT BANKING

Syn Bank Share-of-Wallet Intelligence Engine

A data science and Generative AI approach to estimating client wallet, quantifying Syn Bank's current share, and ranking the highest-priority growth opportunities.

TEAM NAME

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TEAM MEMBERS

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SUBMISSION

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Syn Bank never has the whole client

Corporate clients spread their banking activity across multiple providers, for pricing tension, risk management, and product specialisation. The commercial question is not how much flows through Syn Bank's own books — it is how large the client's **total** wallet is, and where a competitor currently holds the larger share.

1. Quantify capture

What share of each client's total wallet does Syn Bank hold today, across Transactional Banking, Global Markets, and Investment Banking?

2. Rank the gap

Where is the gap between total wallet and current share largest — and most winnable?

3. Apply Gen AI meaningfully

Not a cosmetic extra: a use case that does real analytical work and adds measurable value.

Three internal ledgers, supplemented with public data

What we were given

- Domestic transaction ledger — 2.8m rows
- Cross-border SWIFT payment data
- Trade finance instrument activity
- Validated consistently to **20** JSE-listed clients across all three files, not the 50 the brief described — disclosed openly, not adjusted to fit

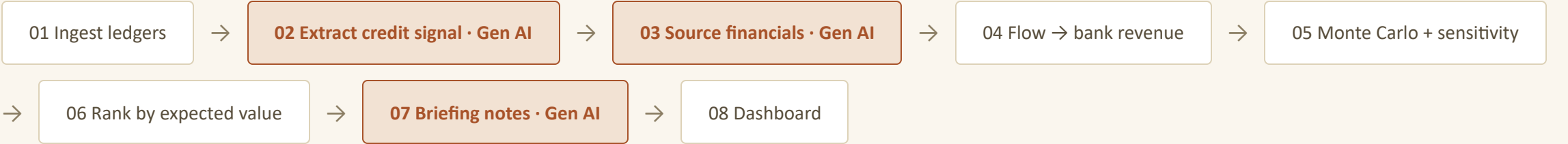
What we sourced externally

- No financial-statement documents were supplied
- Core financials sourced directly from public filings for all 20 clients
- JSE SENS announcements, investor-relations pages, annual reports
- Every external figure cited to its source, with a confidence flag

Where Syn Bank's own ledger goes quiet — no FX hedge, no lending relationship, no trade finance activity — we treat that as a genuine analytical signal, not a data gap to explain away. It usually means a competitor is holding that piece of the relationship.

Every rand traces back to a named assumption

Eight stages, each reading only the previous stage's saved output. The design choice that makes everything downstream defensible: every coefficient that converts observed flow into estimated bank revenue lives in one disclosed, sourced file — never hardcoded inside the model.



TOTAL ADDRESSABLE WALLET

R18.6bn

Across 20 clients, 3 pillars

CAPTURED BY SYN BANK

R977m

Observed from internal ledgers

BLENDED SHARE OF WALLET

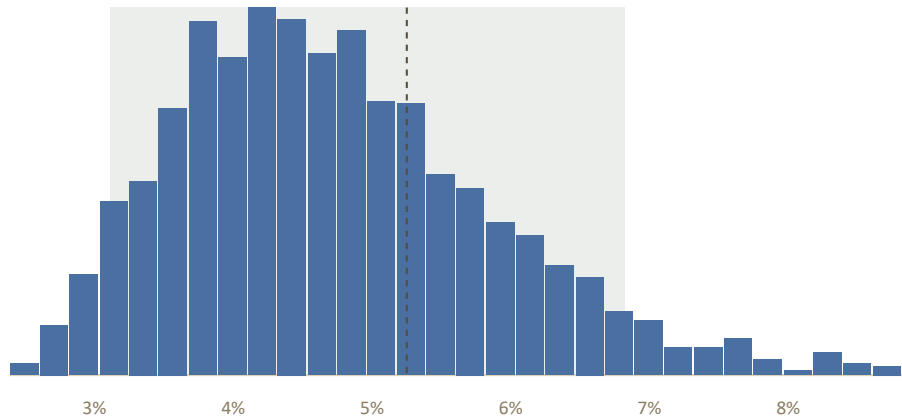
5.3%

Base case — see next slide for range

We report a distribution, not a point estimate

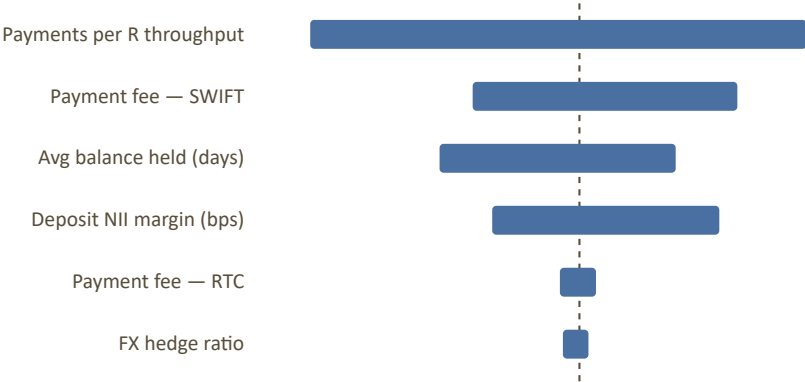
Portfolio blended share of wallet

2,000 Monte Carlo iterations · base case 5.3% · 5th–95th pct 3.1%–6.8%



What the answer is most sensitive to

Swing in blended share as each assumption moves low → high



The single largest driver, by a wide margin, is the least-anchored assumption in the model — payment volume per rand of client throughput. A second, fully independent estimator (product-breadth revealed presence) runs systematically higher for 19 of 20 clients — a disclosed limit of that method on this broad-engagement portfolio, not a contradiction. The one client where it does not diverge, Pepkor, independently confirms a separate finding: our TAM assumption for that client is too conservative.

Three components that do real analytical work

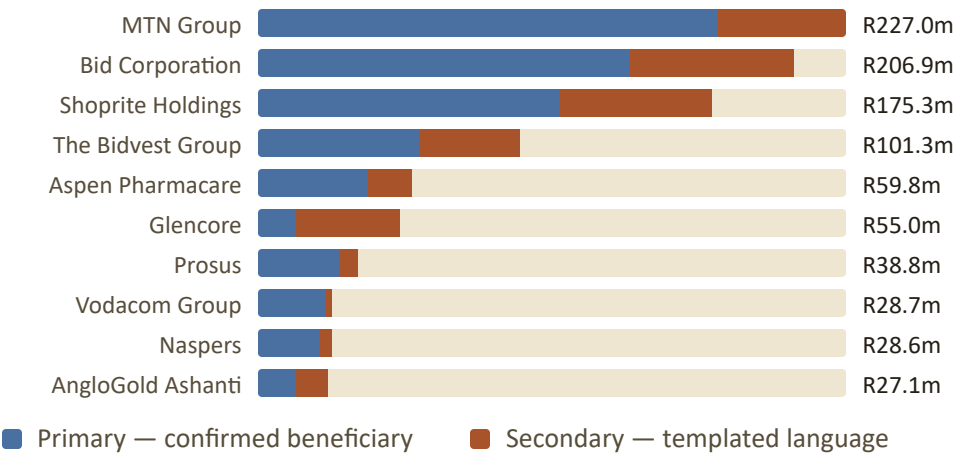
COMPONENT	WHAT IT DOES	HOW WE CHECKED IT
Competitor-credit extraction	Classifies free-text transaction memos to identify credit facilities another bank actually holds	Cross-checked against an independent keyword baseline, row by row
Financial-statement structuring	Reads public filings and extracts revenue, cost of sales, debt, and foreign-revenue exposure to a strict schema	Hand-labelled ground-truth sample, scored field by field
Client briefing notes	Drafts a client-ready summary and recommended next step for every one of the 20 clients	Grounded strictly in already-computed figures — never free-generated

Beyond the brief's minimum: the brief asks for briefing notes on at least 3 clients. We generated all 20 — every prompt logged, every response cached, every call timed and costed.

R1.0bn in flow that belongs to someone else's balance sheet

Primary + secondary tier, by client

Top 10 of 20 clients, R millions



A recurring memo template — *"bridging facility settlement," "syndicate participation settlement"* — appears **4,199** times across all three ledgers. Every occurrence marks a payment Syn Bank is settling for a credit facility another bank originated.

Roughly **2.5×** larger than the scale the original brief implied, because the same fingerprint also appears in the domestic payments ledger.

Why this matters: this list does not depend on a single yield assumption. It is a direct read of the ledger, cross-checked two independent ways. A coverage banker can act on it today.

Ranked by expected value, not raw gap size

wallet_gap × win_probability × margin — with product-sequencing enforced: you win FX off the back of payment flow, not the reverse.

RANK	ENTITY	SPECIFIC PRODUCT	EXPECTED VALUE	WIN PROB.	SEQUENCING
1	Shoprite Holdings	Payment fees (transaction volume & pricing)	R193.8m	54%	Flagged
2	Bid Corporation	Payment fees (transaction volume & pricing)	R168.5m	51%	Flagged
3	MTN Group	Payment fees (transaction volume & pricing)	R141.9m	54%	Flagged
4	Vodacom Group	Payment fees (transaction volume & pricing)	R123.7m	52%	Flagged
5	Sanlam	Payment fees (transaction volume & pricing)	R106.0m	53%	Clear
6	The Bidvest Group	Payment fees (transaction volume & pricing)	R104.0m	54%	Flagged

11 of the top 12 clients' single largest opportunity is the same specific product — payment-fee pricing — not a generic "Transactional Banking" label. 9 of 12 are flagged as not yet ready for a Global Markets or Investment Banking pitch.

What this changes on Monday morning

This week

Open the competitor-credit list. It needs no further validation — only a phone call to the client's treasury team.

Next quarter

Use the ranked opportunity list to set call priorities. Lead with payment-fee pricing conversations where the relationship is ready.

Before pitching cross-sell

Check the sequencing flag first. Nine of twelve clients need a deeper Transactional Banking relationship before an FX or Investment Banking pitch will land.

RANKED OPPORTUNITY VALUE

R1.1bn

Across 12 SA-domestic clients

TOP OPPORTUNITY

Shoprite

R193.8m expected value

SEQUENCING FLAGS

9 / 12

Not yet ready for cross-sell

What we would validate before this goes to a client

Assumption-dependent, disclosed as such

- Every fee, margin, and basis-point figure is an informed market-practice estimate — not Syn Bank's real pricing, which no SA bank publishes
- Win probability is a reasoned, four-signal heuristic — not a calibrated forecast, because no historical win/loss record exists to calibrate against

What we could not verify

- No external benchmark exists to confirm the "correct" share of wallet
- The provided data covers 20 clients, not the 50 described in the brief — a real, unexplained gap we surfaced rather than papered over

The ranking is robust to this uncertainty. The rand values are a disciplined first estimate. Replace the market-practice assumptions with Syn Bank's real pricing, and every figure updates — the ranking logic does not need to change.

Two further questions the brief invited us to explore

Client engagement timing

Do cash cycles suggest a best moment to engage?

Day-level timing showed no usable signal, reported as a null result rather than manufactured. Month-level seasonality did: all four consumer-sector clients in the portfolio peak in December — a real, sector-coherent engagement window.

Latency & cost optimisation

Can model routing reduce Gen AI cost without losing accuracy?

Routed straightforward classification to a smaller model, reserving the larger model for ambiguous cases. Result: 11% cost saved, 100% agreement with the original answers on a held-out sample.

Next step: validate pricing assumptions against Syn Bank's actual rate card, then re-run — the pipeline is fully reproducible end to end from a single command.

THANK YOU

Questions & discussion

Syn Bank Share-of-Wallet Intelligence Engine — built end to end, reproducible from a single command, every assumption and every Gen AI prompt on the record.

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TEAM MEMBERS

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CODE

github.com/juddoooooooooooo/hackathon_aug26